Uber trip analysis in the USA

Project made by:

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Meet Our Team

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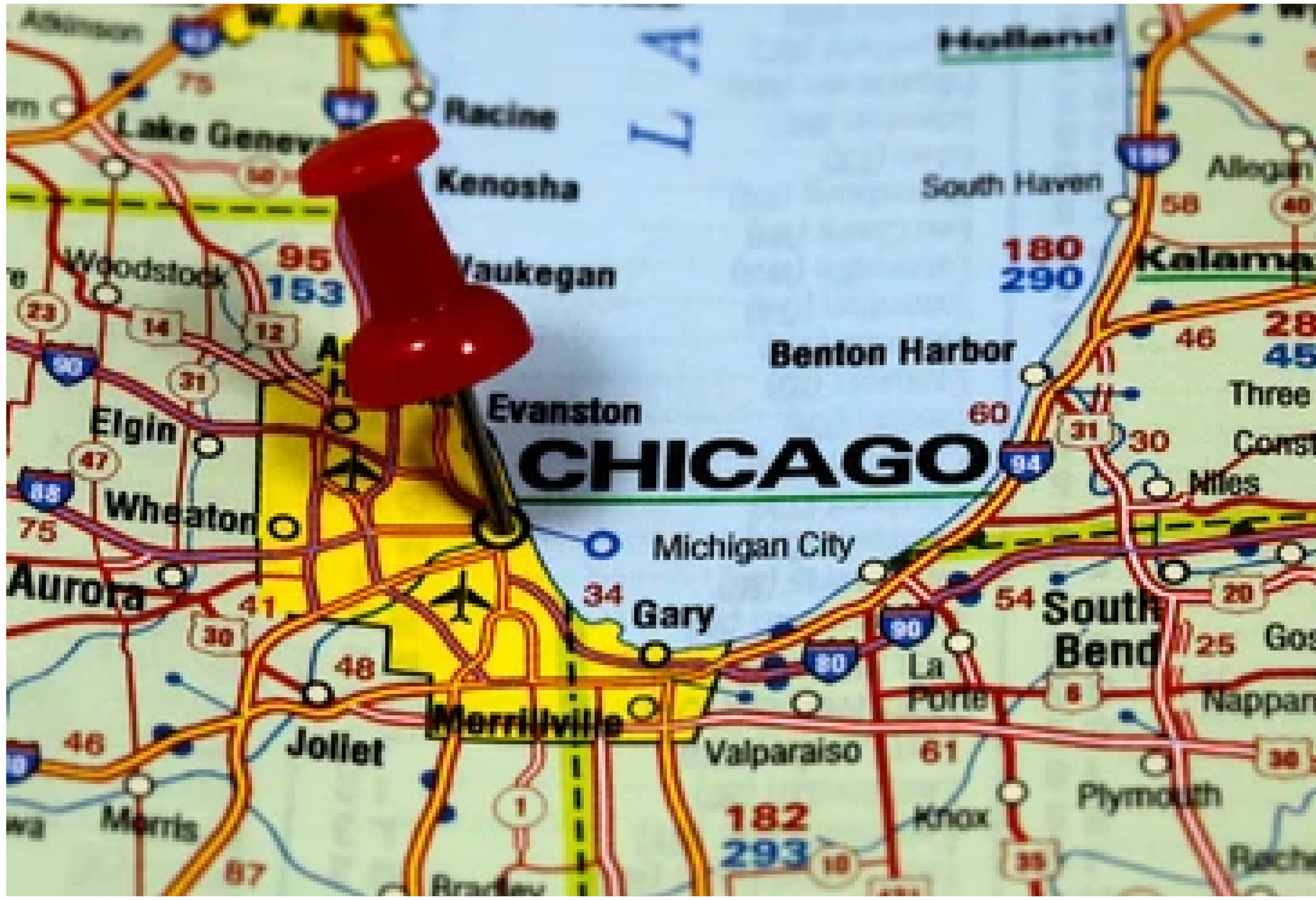
Project concept

The project was based on the analysis of Uber trip data across the United States.

As part of the project, a series of visualizations were developed to illustrate relationships between selected variables and highlight key trends within the dataset.

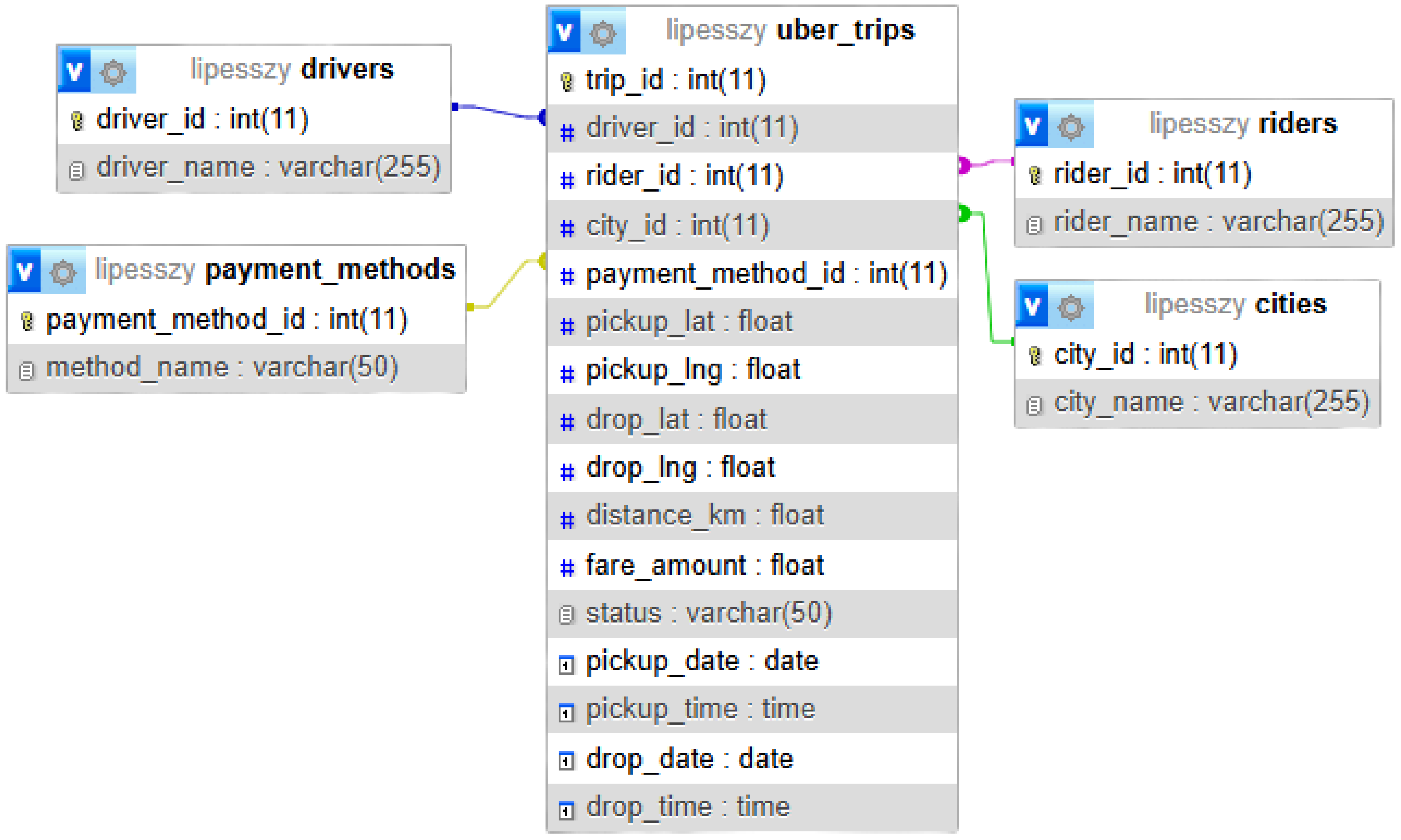
Additionally, interactive filters were implemented to enable dynamic data exploration and allow users to refine the displayed information according to their specific analytical needs, providing deeper insights into the analyzed data.



Uber

Dataset

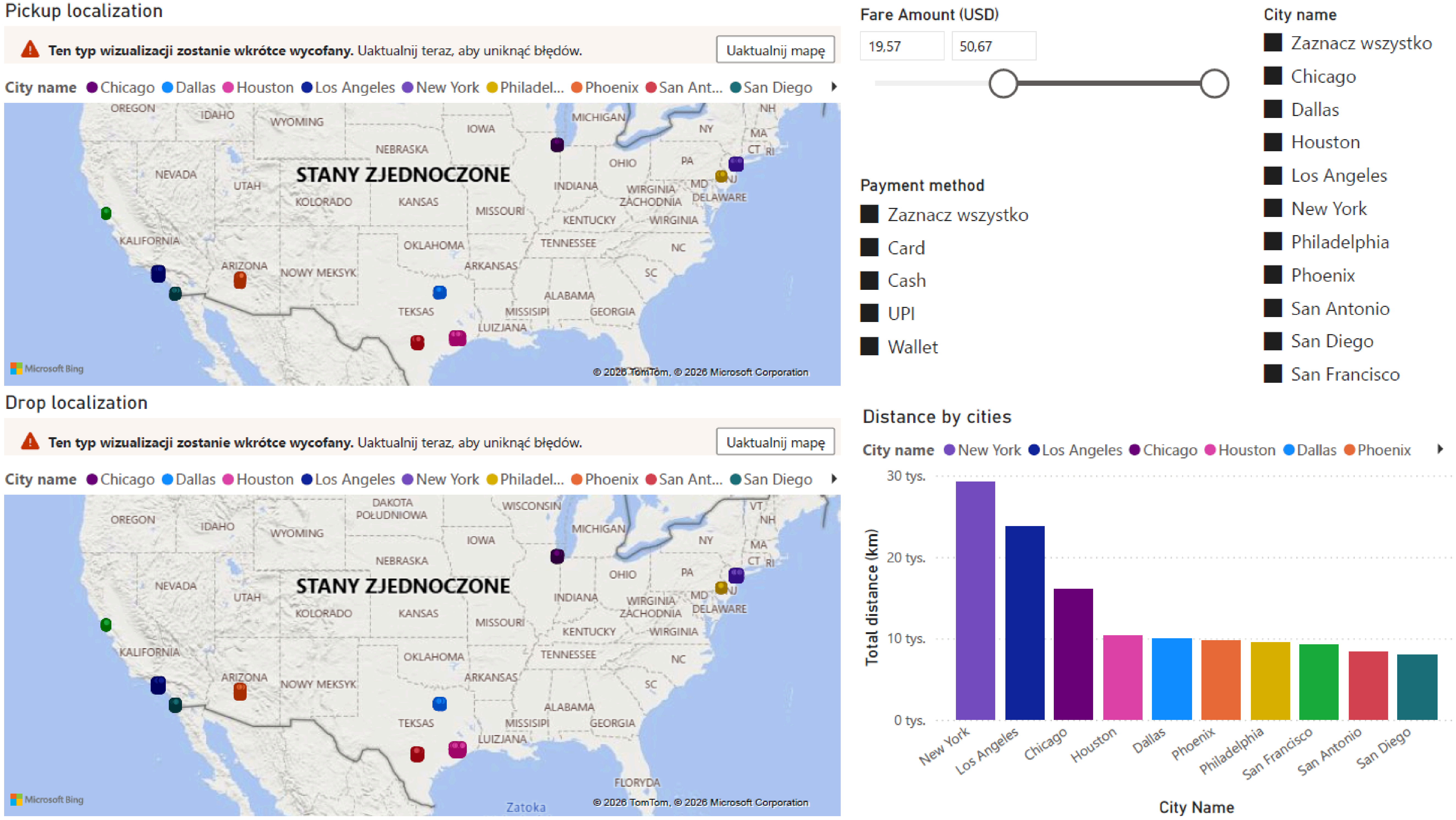
The dataset comprised around 50,000 records of Uber trips conducted across the United States from 1st January till 2nd February 2023. The data were collected from ten major cities: Chicago, Dallas, Houston, Los Angeles, New York, Philadelphia, Phoenix, San Antonio, San Diego, and San Francisco.



Explenation of data

- **trip_id** - the id of the trip
- **driver_id** - the id of the Uber driver
- **driver_name** - Name and Surname of the Uber driver
- **rider_id** - the id of the passager
- **rider_name** - Name and Surname of the passager
- **city_id** - the id of the city
- **city_name** - name of the City
- **payment_method_id** - the id of the payment method
- **method_name** - name of the payment method (Card/Wallet/UPI/Cash)
- **status** - status of the ride (Cancelled/Completed/No-show)
- **distance_km** - distance of the ride
- **fare_amount** – the amount of the fare (the price paid for a trip or ride)
- **pickup_date** – the date when the passenger was picked up
- **pickup_time** – the time when the passenger was picked up
- **pickup_lat** – the latitude of the pickup location
- **pickup_lng** – the longitude of the pickup location
- **drop_date** – the date when the passenger was dropped off
- **drop_time** – the time when the passenger was dropped off
- **drop_lat** – the latitude of the drop-off location
- **drop_lng** – the longitude of the drop-off location

Visualisation, by the localization



Our database allowed us to analyse Uber journeys in the biggest US cities, taking into consideration the characteristics of said transfers such as:

- city of origin
- fare amount
- payment method
- travelled distance
- pickup and drop localization

Said data is sufficient to perform different market analysis operations.

Time Series Analysis

Our work allows tracking demand for the service and customer behhaviour in the context of a chosen time period .

Status Classification

Trips are segmented into distinct operational outcomes: Completed, Cancelled, and No-show.

Temporal Overview - January

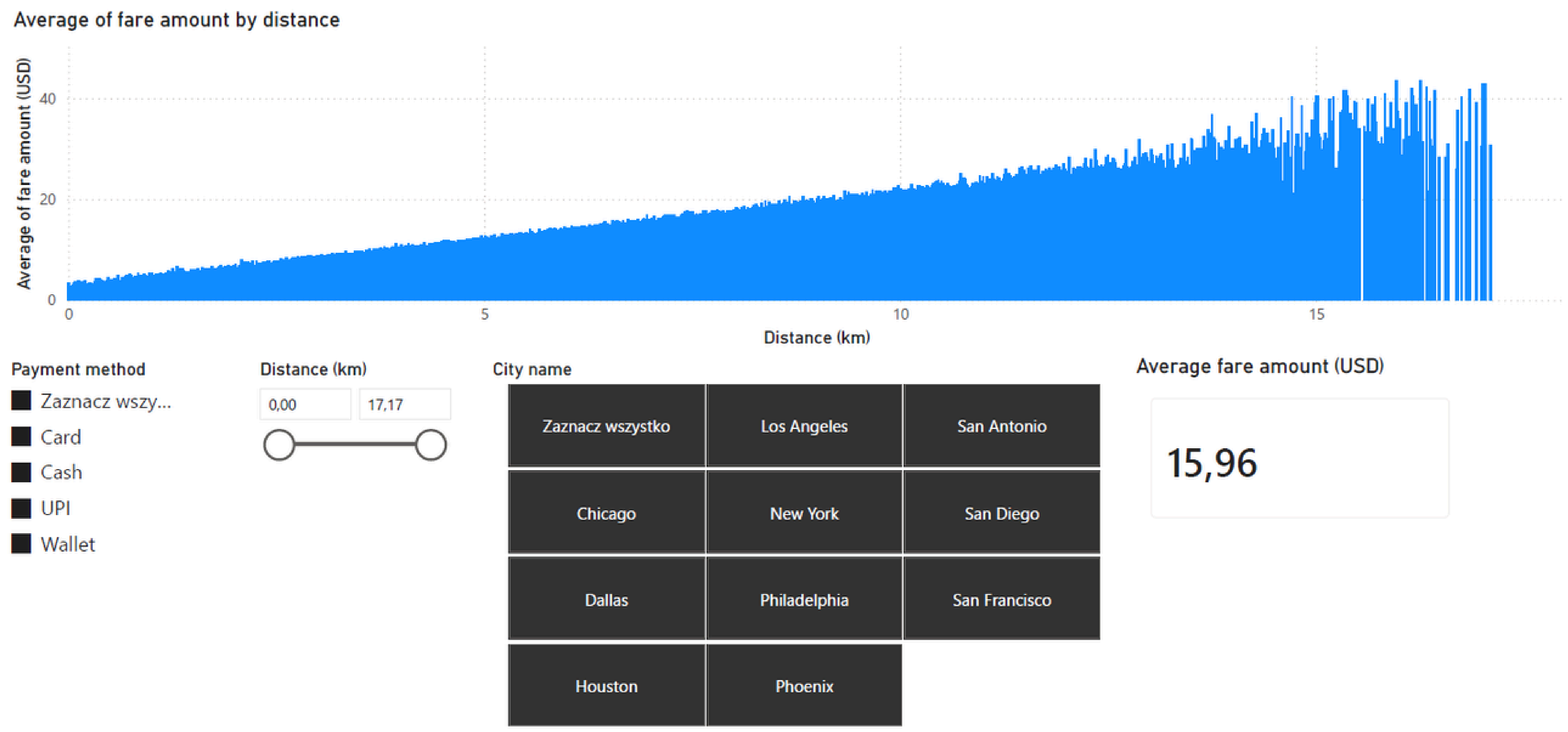
The shown figure allows for tracking the number of Uber trips in relation to two metrics:

- day of the month
- trip status

Practical implications

This visualization provides valuable insight into how customer behaviour varies during the month. It tracks how the numer of trips of any given status varies in correspondence to holidays etc.

Fare vs. Distance Correlation



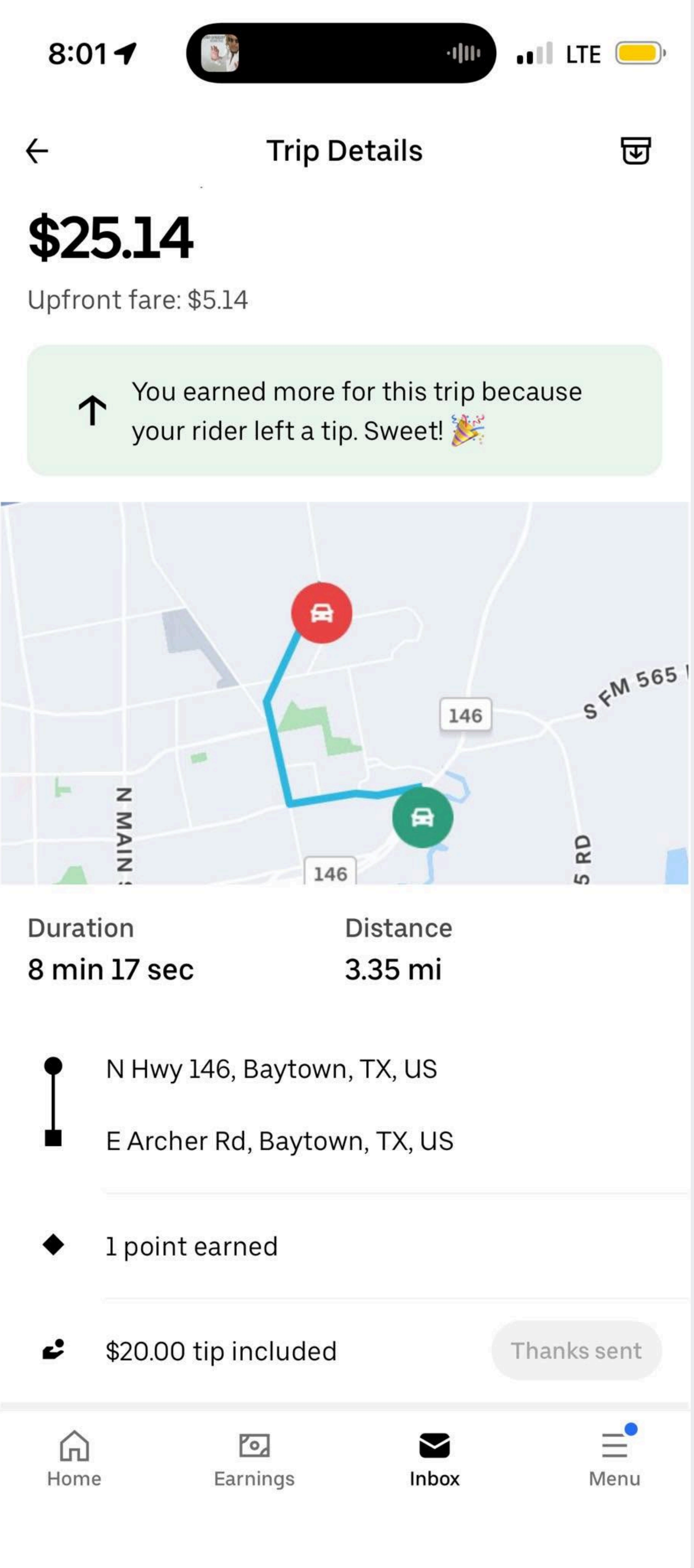
- We can analyse the cost of a fare based on its length, with regard to the city of origin.
- Our visualization allows for the analysis of multiple cities at once, which lets us determine the local as well as global statistics of Uber transportation prices in the US.
- We are also able to examine customer behaviour, analysing the preferred payment methods in relation to the city of origin and trip distance.

Interesting statistics

Payment method	Average fare amount for all Cities	Highest/Lowest	City name	Average amount by city
All methods	$15.98	Lowest	Houston	$15.89
		Highest	Dallas	$16.20
Cash	$15.97	Lowest	Houston	$15.77
		Highest	Chicago	$16.18
UPI	$15.95	Lowest	San Diego	$15.67
		Highest	Dallas	$16.29
Wallet	$15.97	Lowest	Philadelphia	$15.61
		Highest	San Diego	$16.17
Card	$16.02	Lowest	Philadelphia	$15.76
		Highest	Dallas	$16.37

Interesting statistics

City name	Total distance (km)	Average distance (km)
New York	76714,78	7,00
Los Angeles	62172,16	6,98
Chicago	42190,09	7,03
Houston	27758,25	6,97
Philadelphia	25009,42	7,01
Phoenix	24953,83	7,01
Dallas	24636,19	7,09
San Francisco	24286,32	6,98
San Antonio	21643,81	7,01
San Diego	21038,66	7,05


8:01
LTE
Trip Details
$25.14
Upfront fare: $5.14
You earned more for this trip because your rider left a tip. Sweet! 🎉
146
S FM 565
146
Duration
8 min 17 sec
Distance
3.35 mi
N Hwy 146, Baytown, TX, US
E Archer Rd, Baytown, TX, US
1 point earned
$20.00 tip included
Thanks sent
Home
Earnings
Inbox
Menu

Thank You